Java API for XML Processing 1.1 What's New

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Introduction

- JAXP 1.0 emerged to fill deficiencies in existing industry standards: SAX 1.0 and DOM Level 1.
- Examples:
  - Bootstrapping a DOM tree
  - Controlling parser validation
- Since JAXP 1.0:
  - Industry standards changed: SAX 2.0, DOM Level 2
  - JAXP expanded to satisfy more needs: XSLT

JAXP 1.1

- JAXP enables apps to parse and transform XML documents
- Allows apps to be independent of a particular implementation
- Augments existing SAX and DOM API standards

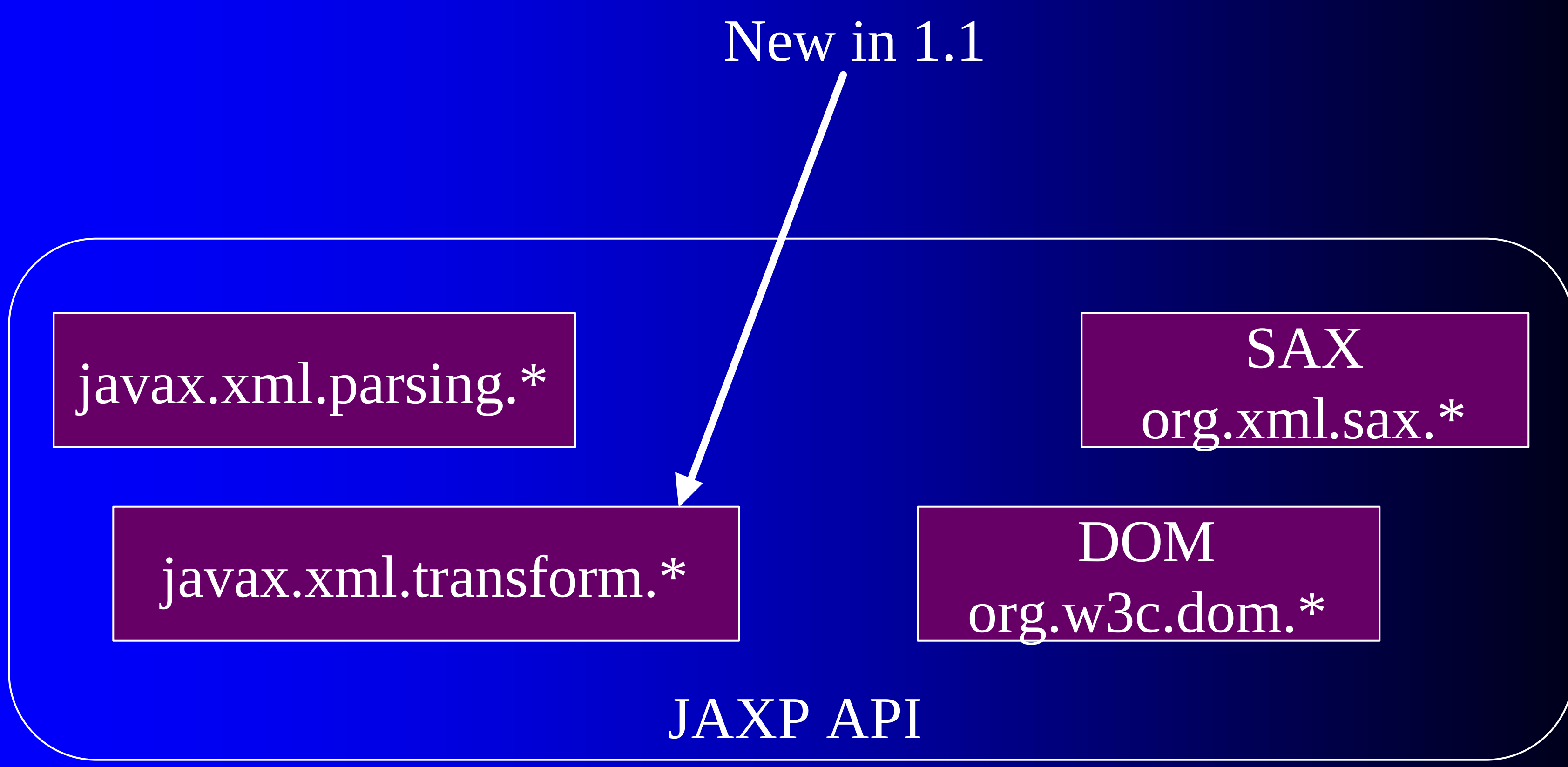
New Features in 1.1

- Support for XSLT 1.0 based on TrAX (Transformation API for XML)
- Name change from “Parsing” to “Processing”
- Parsing API updated to SAX 2.0 and DOM Level 2
- Improved scheme to locate pluggable implementations

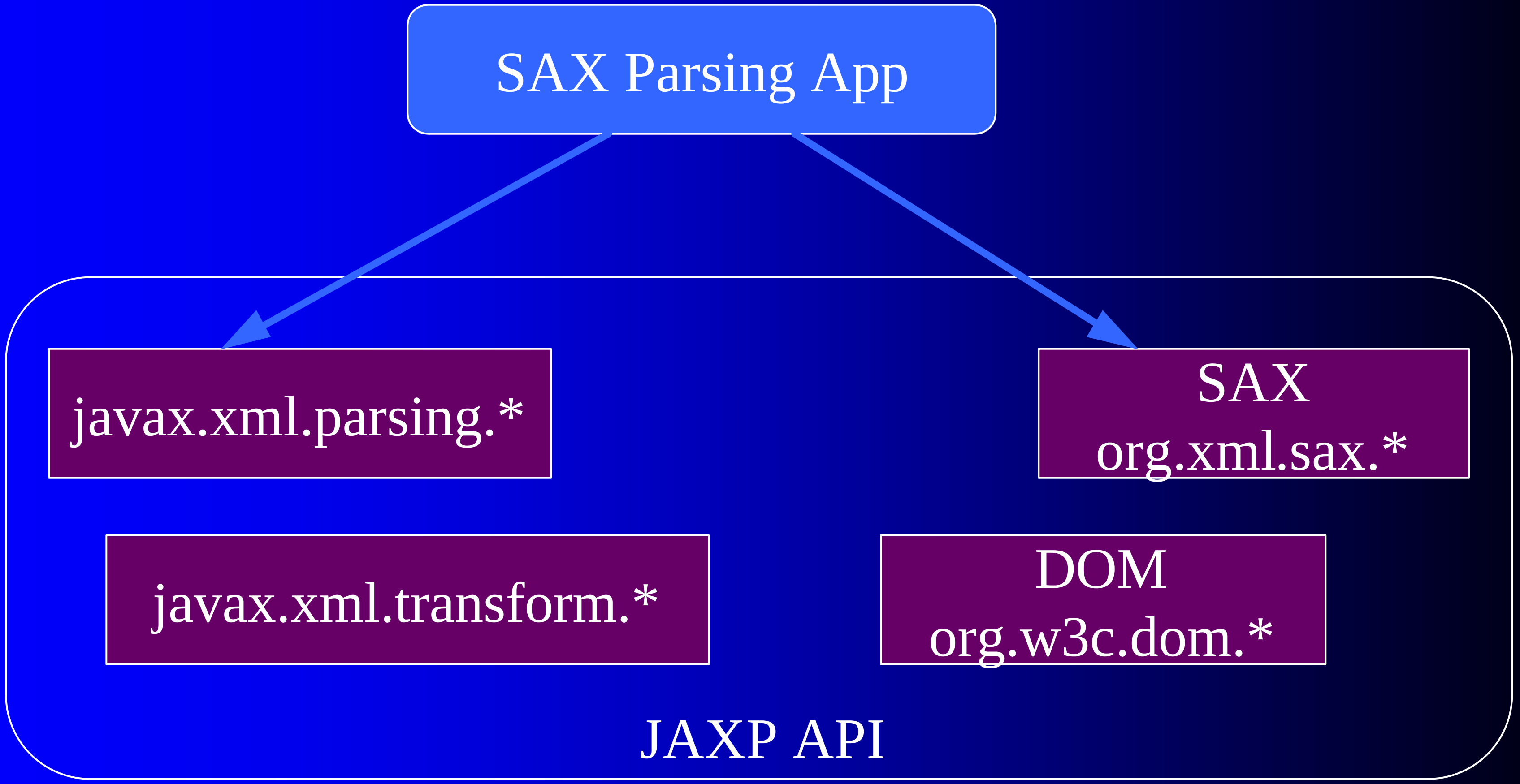
Application Usage Categories

- Parsing using SAX 2.0
- Parsing using DOM Level 2
- Transformation using XSLT

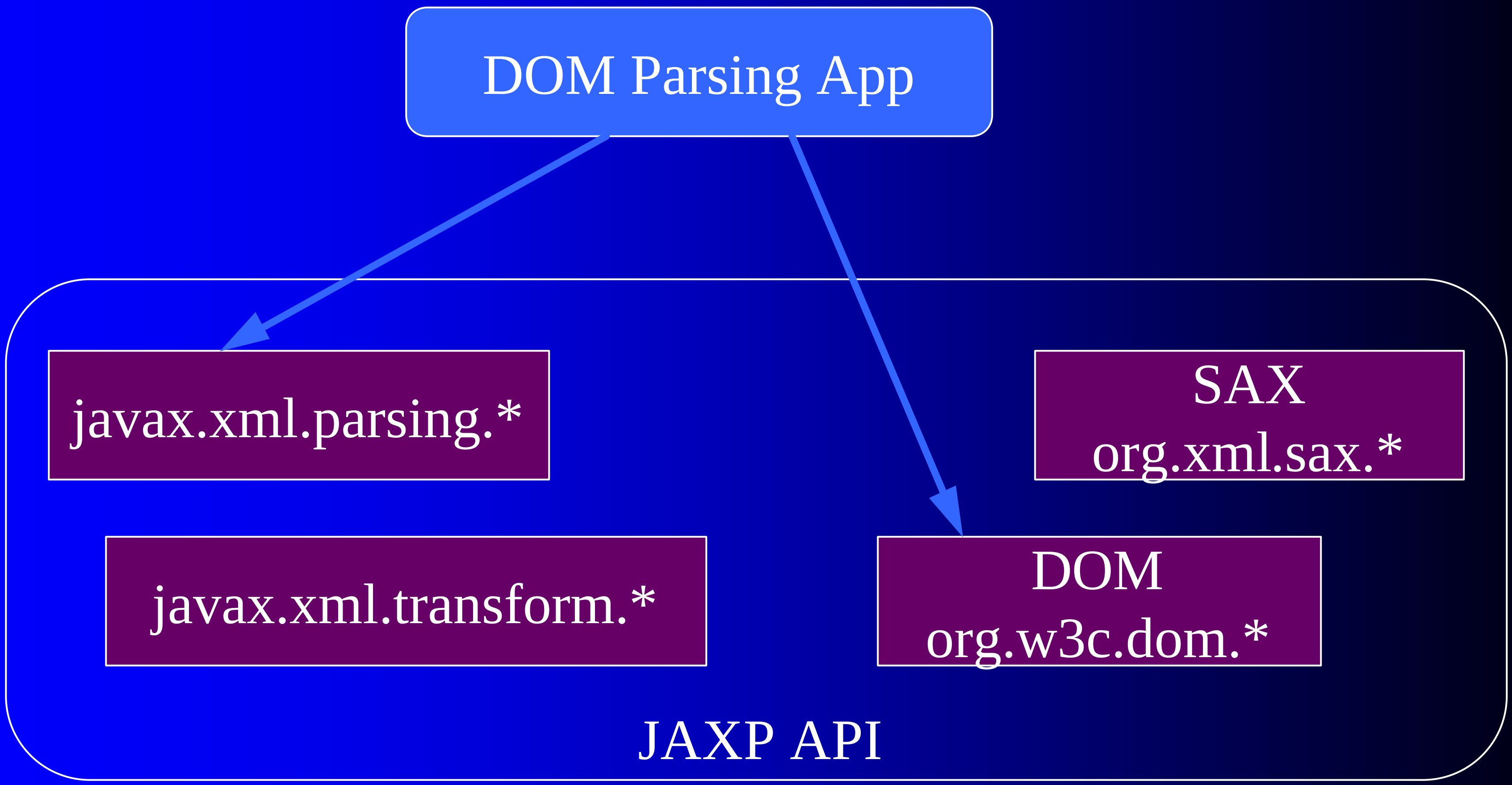
JAXP 1.1 Components



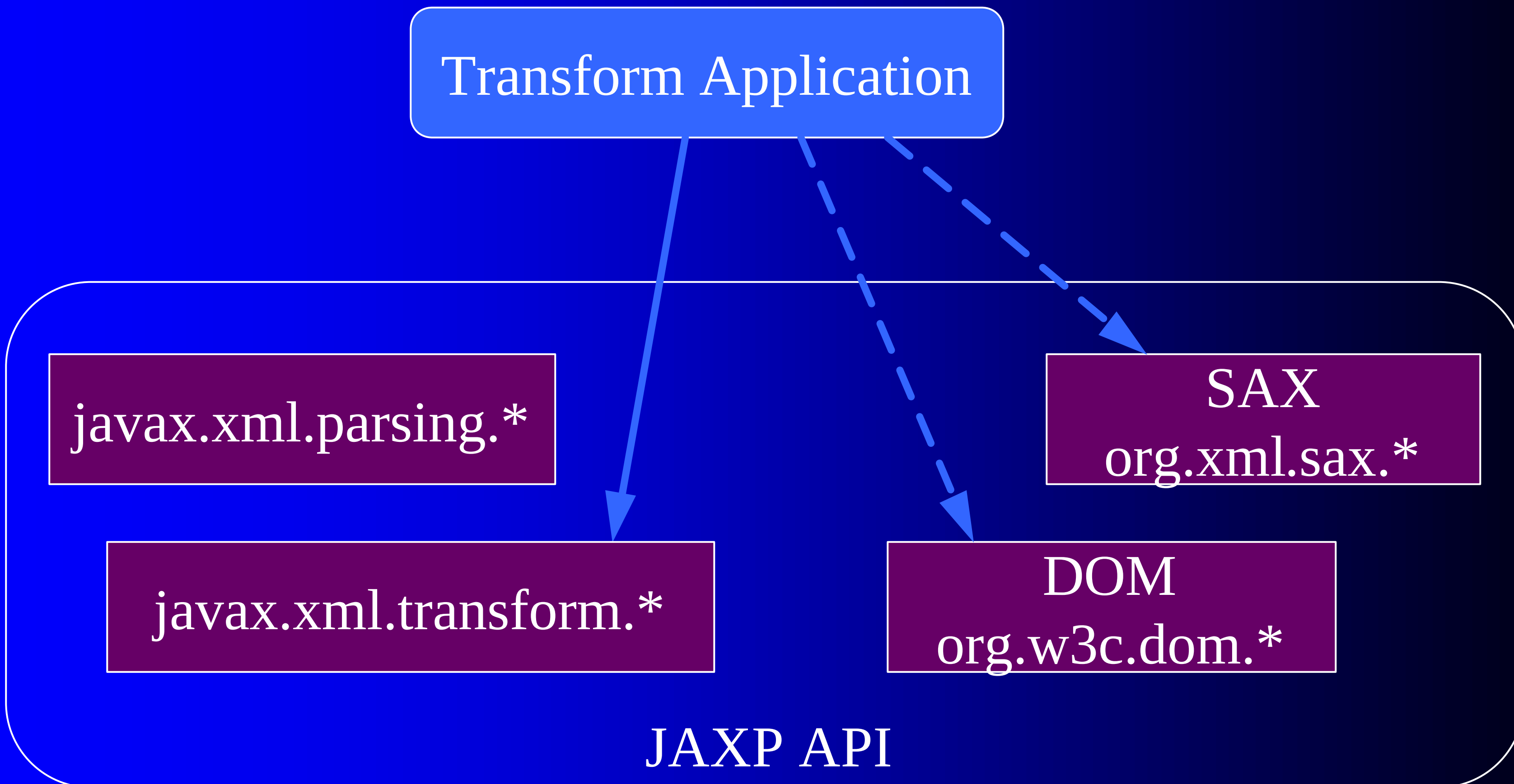
1) SAX Parsing Application



2) DOM Parsing Application



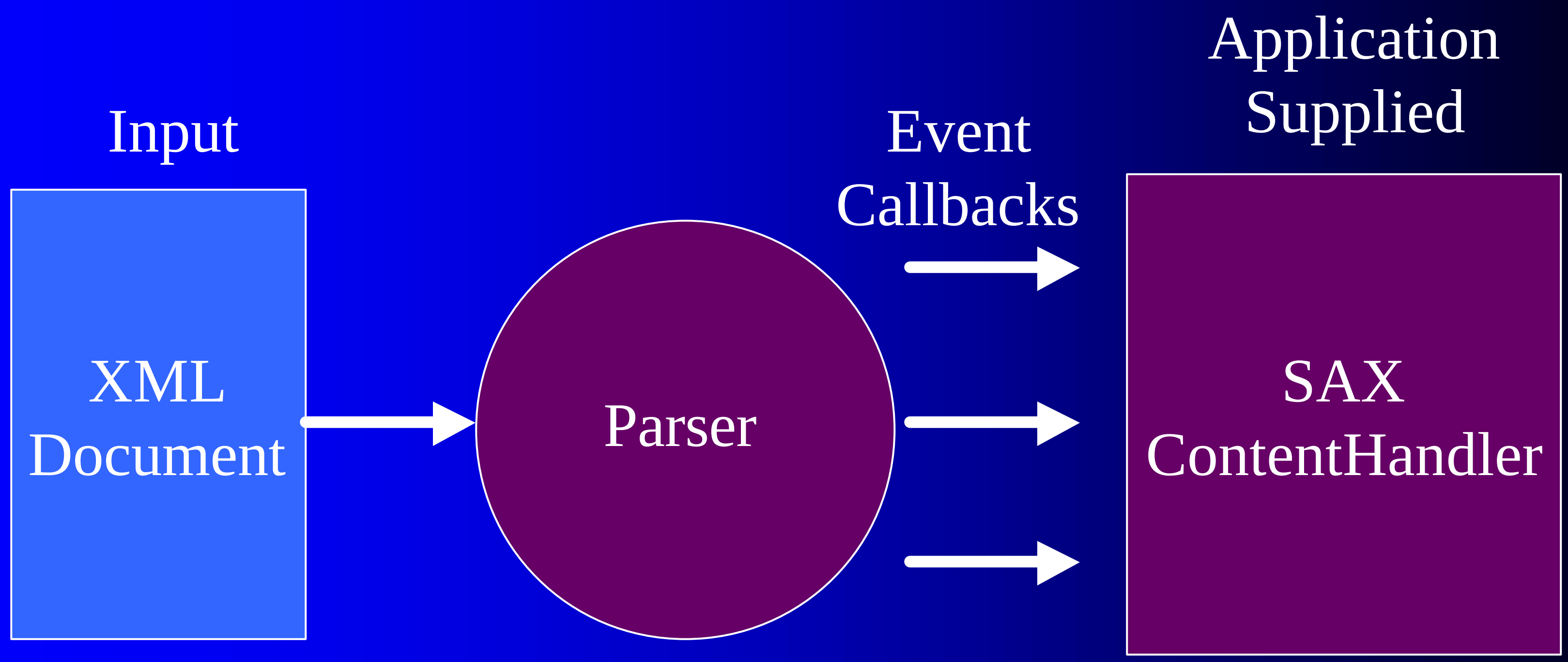
3) Transform Application



1) Parsing using SAX 2.0

- Application supplies a SAX ContentHandler to parser
- Application tells parser to start parsing a document
- Parser calls methods in the ContentHandler that application previously supplied during parse

SAX 2.0 Parsing



SAX ContentHandler

```
public interface ContentHandler {
  void startElement(namespaceURI,
          localName, qName, atts);
  void endElement(namespaceURI,
          localName, qName);
  void characters(ch[], start, length);
  ...
}
```

Example: SAX2 Application

```
1. XMLReader xmlReader = create SAX2 XMLReader
                         instance

2. xmlReader.setContentHandler(myContentHandler);

3. xmlReader.parse(myInputSource);
```

Create XMLReader

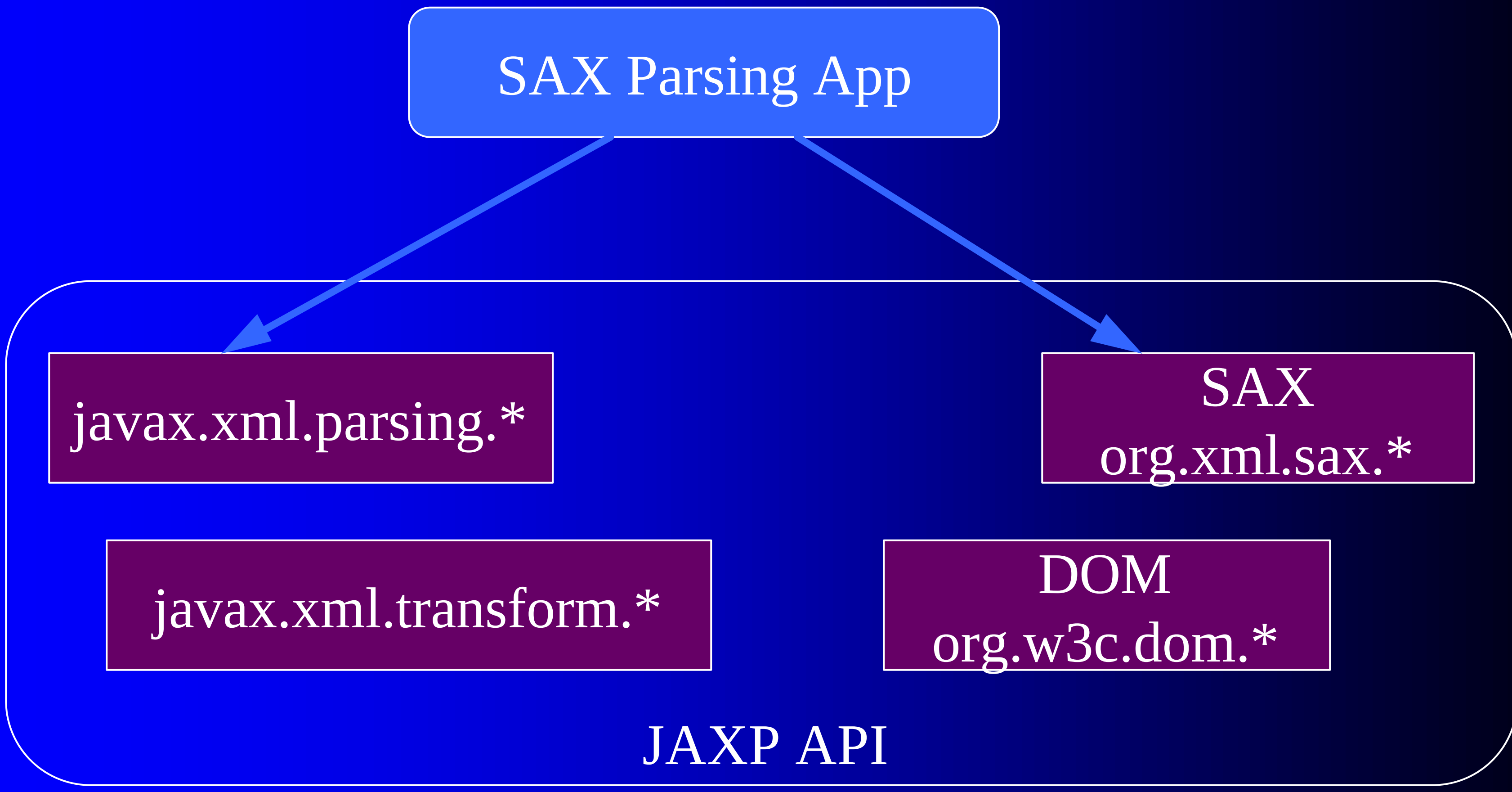
```
SAXParserFactory spf =
  SAXParserFactory.newInstance();

// Change namespaces feature to SAX2 default
spf.setNamespaceAware(true);

// Create SAX XMLReader w/ SAX2 default features
XMLReader xmlReader =
  spf.newSAXParser().getXMLReader();

// Use xmlReader as you would normally
...
```

SAX Parsing Application



Changing the Implementation

- Define a system property: `javax.xml.parsers.SAXParserFactory`
- $JAVA_HOME/jre/lib/jaxp.properties file
- Jar Service Provider

  `META-INF/services/javax.xml.parsers.SAXParserFactory`
- Platform default (fallback)

Jar Service Provider

- Jar file may contain a resource file called …/javax.xml.parsers.SAXParserFactory containing the name of a concrete class to instantiate
- JAXP static SAXParserFactory.newInstance() method searches classpath for resource and instantiates specified concrete class

Examples: Using a Particular Implementation

With Java 2 version 1.3

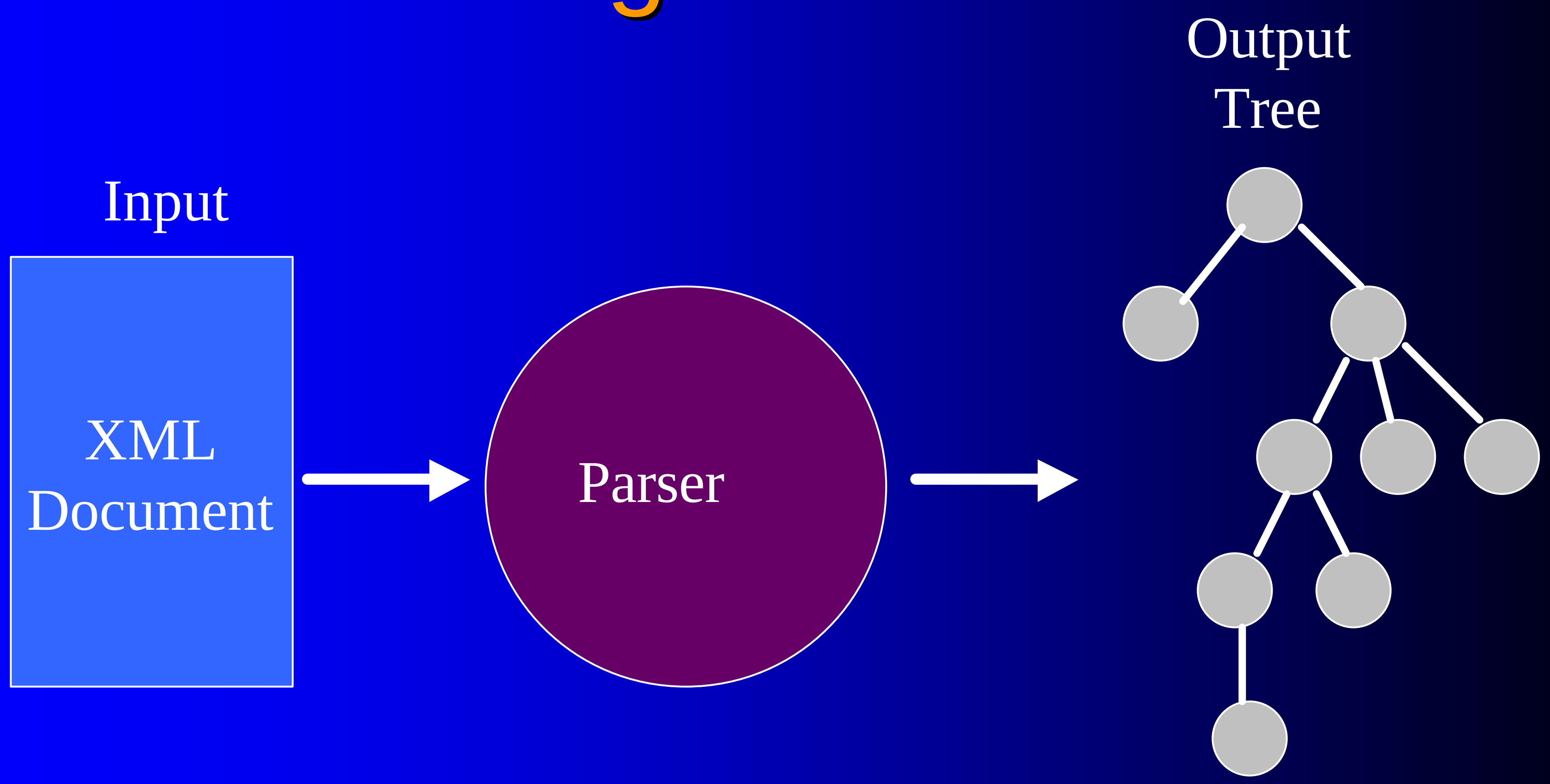
- To use Xerces, use classpath =
  - xerces.jar (contains all classes)
- To use Crimson, use classpath =
  - jaxp.jar (contains javax.xml.*)
  - crimson.jar (contains sax, dom)
- You get Xerces, if classpath =
  - jaxp.jar
  - xerces.jar
  - crimson.jar

(Jar file names correct as of Feb 2001)

2) DOM Parsing Example

- Application gives DOM builder an XML document to parse
- Builder returns with a DOM Document object representing the DOM “tree”

DOM Parsing



JAXP Adds to DOM Level 2

- Method to “Load” a DOM Document object from an XML document*
- Methods to control parser behavior such as validation and error handling*
- Provides pluggable DOM parser implementation

* Proposed for Level 3

JAXP DOM Example

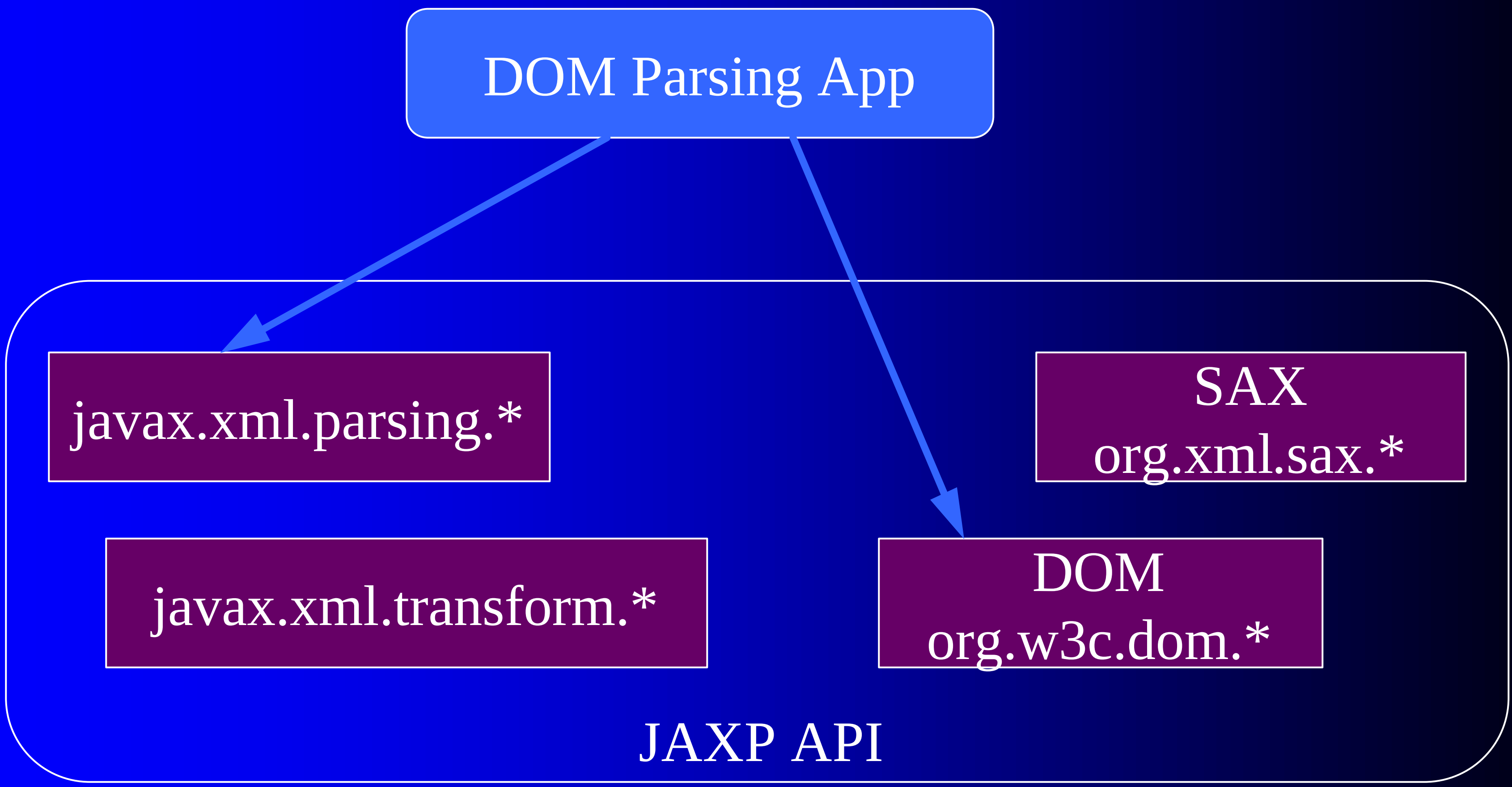
```
// Get platform default implementation
DocumentBuilderFactory dbf =
  DocumentBuilderFactory.newInstance();
// Set options
dbf.setNamespaceAware(true);
dbf.setValidating(true);

// Create new builder
DocumentBuilder db = dbf.newDocumentBuilder();

db.setErrorHandler(myErrorHandler);
Document doc =
  db.parse(“http://server.com/foo.xml”);

// Use doc with usual DOM methods
...
```

DOM Parsing Application



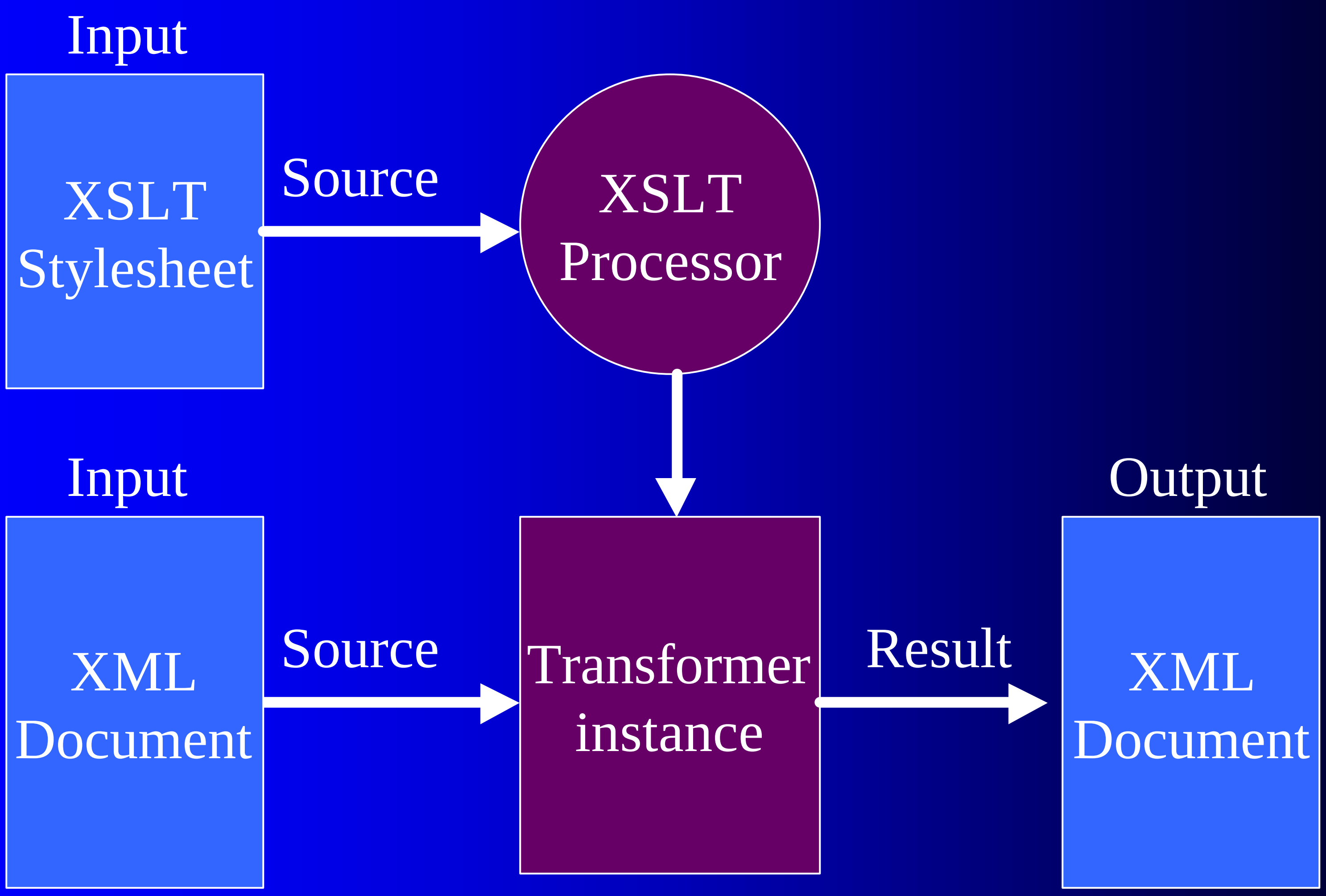
JAXP DocumentBuilderFactory

- Has same pluggability mechanism as SAXParserFactory does

3) Transformation Using XSLT

- Major new feature
- Enables transformation of one XML document into another XML document using XSLT 1.0
- API based on TrAX, Transformation API for XML, initiated by Scott Boag, Michael Kay, and others. Incorporated into JAXP version 1.1.
- Provides pluggable Transform implementation similar to parsing

Transformation Diagram



Transformation Example

- Create `Transform` instance from XSLT stylesheet
- Use the `Transform` instance to transform the source document into a result document by calling:

```
Transform.transform(Source, Result)
```

Transform Arguments

- Specialized implementations of generic `Source` and `Result` interfaces are in
  - javax.xml.transform.stream
  - javax.xml.transform.sax
  - javax.xml.transform.dom
- Different combinations of `Source` and `Result` can be passed to `transform()` method
- Examples: StreamSource, DOMSource, SAXResult, StreamResult

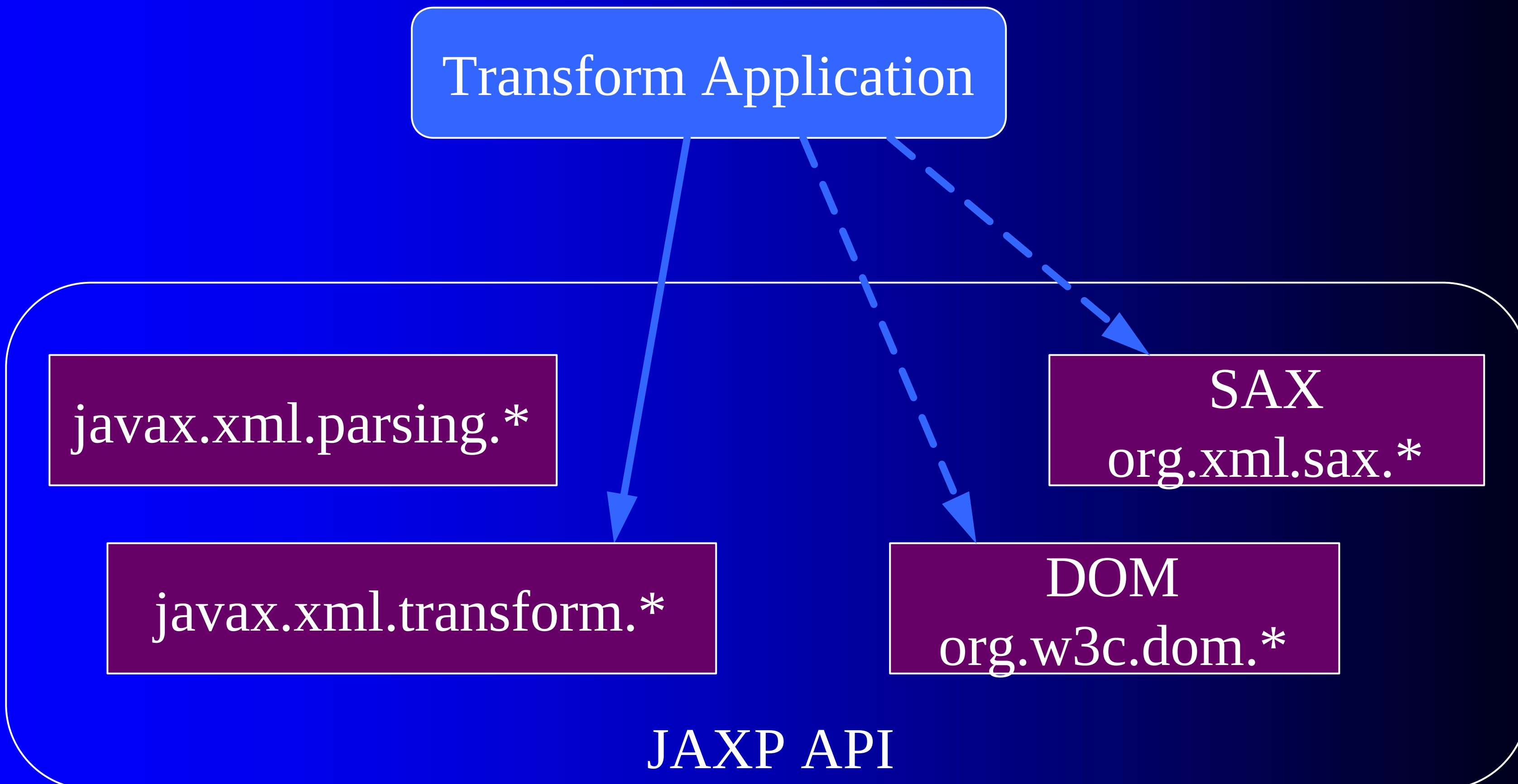
Transform Example Code

```
// Get concrete implementation
TransformFactory tf =
  TransformFactory.newInstance();

// Create a transformer for a particular stylesheet
Transformer transformer = tf.newTransformer(
    new StreamSource(stylesheet));

// Transform input XML doc to System.out
transformer.transform(new StreamSource(sourceId),
    new StreamResult(System.out));
```

Transform Application



Example XSLT Stylesheet

```
<?xml version="1.0"?>
<xsl:stylesheet
    xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
    version="1.0">
  <xsl:template match="/">
    <html>
      <head>
        <title>Stock Quotes</title>
        <style type="text/css">
          <xsl:comment>
            body { background-color: white; }
          </xsl:comment>
        </style>
      </head>
```

Example XSLT Stylesheet (cont)

```
      <body>
        <center>
          <h1>Stock Quotes</h1>
        </center>
        <xsl:apply-templates/>
        <img src="{$image-name}"/>
      </body>
    </html>
  </xsl:template>
</xsl:stylesheet>
```

Example: Output a DOM Tree

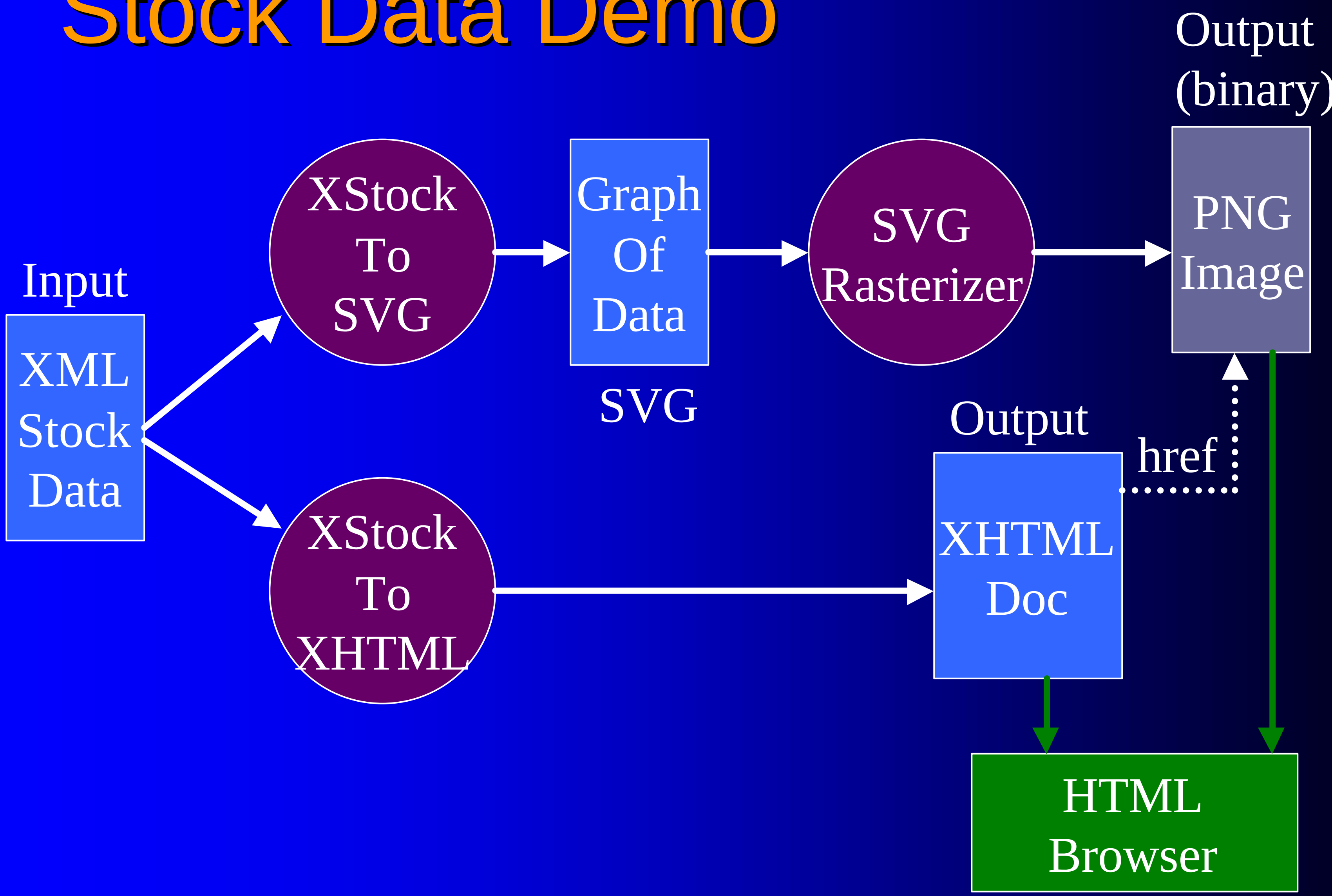
- How do I output a DOM tree as XML?
- Future: a requirement of DOM Level 3
- Can use a JAXP 1.1 transform to do this
  - Create an identity Transformer
  - Transform DOMSource to StreamResult:

```
identity.transform(domSource, streamResult)
```

Demonstration

- Transform stock data in some XML format into a document which can be viewed in an HTML browser.
- Use 2 transforms:
  - Stock data to SVG
  - Stock data to XHTML
- Rasterize SVG into image

Stock Data Demo



References

- My web page: www.apache.org/~edwingo
- Apache XML projects: xml.apache.org
- SAX: www.megginson.com/SAX
- DOM: www.w3c.org/DOM
- JAXP specs, RI: java.sun.com/xml
- SAXON: users.iclway.co.uk/mhkay/saxon

Questions